



Plant-level yield prediction of quinoa in field plots across growth stages using UAV-derived 3D structural and spectral features with machine learning

Sayo Sesay^a, Tomoaki Yamaguchi^b, Sayako Kushino^a, Yasuharu Yoshikawa^c, Shunsuke Adachi^c, Keisuke Katsura^{d,*}

^a United Graduate School of Agricultural Science, Tokyo University of Agriculture and Technology, Fuchu, Tokyo, 183-8509, Japan

^b Faculty of Applied Biological Sciences, Gifu University, Yanagido, Gifu, 501-1193, Japan

^c Graduate School of Agriculture, Tokyo University of Agriculture and Technology, Fuchu, Tokyo, 183-8509, Japan

^d Graduate School of Agriculture, Kyoto University, Kitashirakawa-Oiwake-cho, Sakyo, Kyoto, 606-8502, Japan

ARTICLE INFO

Keywords:

Quinoa
UAV remote sensing
Yield prediction
Machine learning
3D
Growth stage
Precision agriculture

ABSTRACT

Quinoa (*Chenopodium quinoa* Willd.) is a nutritionally important pseudo-cereal with high environmental adaptability, but its yield potential remains underexploited due to limited precision monitoring tools. Current yield assessment relies on destructive sampling, which limits spatial and temporal resolution. While UAV-based remote sensing is extensively used for cereal yield prediction, quinoa applications remain largely limited to process-based modeling, representing a critical knowledge gap. This study presents the first demonstration of UAV-based, plant-level yield prediction for quinoa using machine learning and 3D structural features. Unlike major cereals, quinoa exhibits highly diverse canopy colors and complex morphology across genotypes and growth stages, making spectral vegetation indices unreliable. Therefore, 3D structural features provide a more robust basis for yield prediction by integrating branching complexity and canopy density, which directly relate to biomass accumulation and reproductive capacity. We conducted UAV surveys combined with machine learning across five growth stages over two seasons (2022 & 2024–2025). RGB and multispectral imagery generated 16 features, including 3D structural parameters and vegetation indices. Five ML regression models were evaluated at each stage, with Random Forest at flowering achieving optimal accuracy ($R^2 = 0.726$, $RMSE = 16.71 \text{ g/m}^2$). Early and late vegetative stages demonstrated slightly lower but still robust prediction accuracy ($R^2 = 0.690$ – 0.704), enabling practical early-season management decisions. Notably, 3D structural features, particularly plant volume and surface area, substantially outperformed spectral indices. These results validate UAV-based ML for non-destructive quinoa yield prediction and demonstrate the essential role of 3D structural features for architecturally complex crops.

1. Introduction

Quinoa (*Chenopodium quinoa* Willd.) has emerged as a critical crop for global food security due to its exceptional nutritional composition and remarkable environmental adaptability, particularly in the face of climate change and increasing population demands [1,2]. Global production has expanded substantially over the past two decades, with Bolivia and Peru maintaining their position as dominant producers, accounting for the majority of worldwide quinoa cultivation [3]. This growth trajectory reflects considerable economic potential, with

increasing market valuations driven by growing consumer demand for nutritious, gluten-free alternatives and expanding cultivation into new geographic regions [4,5]. As cultivation extends beyond traditional Andean regions to address global food security challenges, improved agricultural management practices and accurate yield prediction systems are essential for optimizing resource allocation and management decisions.

Traditional yield estimation methods, often relying on destructive sampling or subjective visual assessments, are labor-intensive, time-consuming, and provide limited spatial coverage, making them

* Corresponding author.

E-mail address: katsura.keisuke.2a@kyoto-u.ac.jp (K. Katsura).

<https://doi.org/10.1016/j.atech.2026.101940>

Received 17 November 2025; Received in revised form 9 February 2026; Accepted 28 February 2026

Available online 28 February 2026

2772-3755/© 2026 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

inadequate for modern precision agriculture applications [6]. These limitations become particularly critical as quinoa cultivation expands into diverse agroecological zones with novel stress patterns, where traditional management knowledge may prove insufficient [7–9].

Quinoa presents distinct advantages for developing advanced monitoring systems that address these limitations. Individual plants exhibit substantial within-field yield variability driven by localized environmental factors, while morphological traits demonstrate strong correlations with final grain production [10–12]. Additionally, traditional wide-spacing cultivation practices [7] create distinct, non-overlapping plant canopies that facilitate accurate segmentation and individual plant identification from aerial imagery, enabling precise plant-level monitoring and assessment. These characteristics make quinoa particularly well-suited for unmanned aerial vehicle (UAV)-based remote sensing approaches.

The integration of UAVs with advanced imaging sensors has revolutionized crop monitoring and yield prediction capabilities for major crops such as wheat, rice, and maize [13–15]. UAV-based remote sensing offers several advantages over traditional methods and satellite imagery, including high spatial and temporal resolution, flexible data acquisition schedules, and cost-effective monitoring of field-scale experiments [16,17]. Quinoa's indeterminate growth habit and variable branching patterns create a complex canopy architecture that is particularly well-suited to UAV-based approaches. Rather than relying solely on two-dimensional (2D) spectral information, UAVs equipped with photogrammetry capabilities can capture detailed three-dimensional (3D) structural data that quantifies actual canopy physical structure throughout development. These 3D features are especially valuable for quinoa, as they can capture the vertical stratification and lateral branching complexity that characterize its architecture-structural variations that 2D spectral indices may fail to adequately represent. By directly measuring canopy volume, surface area, and height dynamics, 3D structural features provide insights into biomass accumulation and architectural development that are directly relevant to yield, overcoming limitations of spectral indices, which may saturate or be less sensitive to structural variations in complex canopies [18–20].

Advances in machine learning (ML) over the past decade have significantly enhanced the potential for accurate crop yield prediction from UAV-derived data. Machine learning algorithms excel at extracting complex, non-linear relationships from high-dimensional remote sensing datasets, offering superior performance compared to traditional parametric models in crop yield prediction applications [21]. These algorithms automatically discover patterns within remote sensing data, adapting to the inherent complexity of crop-environment interactions [22,23]. The combination of high-resolution UAV imagery and ML algorithms is particularly promising for capturing the complex relationships between plant structure, spectral characteristics, and final yield in crops with variable architecture.

While UAV-based remote sensing has been extensively applied to major cereal crops, emerging research on quinoa demonstrates the potential of these technologies for specialty crop monitoring. Recent studies have begun exploring UAV applications for quinoa phenotyping and management, though with a focus distinct from yield prediction. Jiang et al. [24] successfully employed UAV-retrieved leaf area index (LAI) and SPAD-based chlorophyll measurements combined with random forest modeling to phenotype a quinoa diversity panel, demonstrating the crop's amenability to high-throughput remote sensing approaches. Sankaran et al. [25] utilized UAV-based multispectral imaging for high-throughput field phenotyping to assess irrigation treatment effects in quinoa, highlighting the technology's utility for evaluating management practices. Most recently, Sesay et al. [20] demonstrated that fusion of UAV-based 3D mesh and spectral features improved quinoa biomass and LAI estimation across genotypic and temporal variations, providing compelling evidence that integrated structural-spectral approaches can capture quinoa's morphological

complexity. Additionally, Ivushkin et al. [26] applied UAV technology for soil salinity assessment in quinoa cropland, addressing a critical environmental stress factor. While these studies have advanced UAV applications for quinoa monitoring, assessment of abiotic stress responses, and trait phenotyping, none have directly addressed the critical challenge of yield prediction—a fundamental requirement for precision agriculture implementation and management decision-making.

The absence of UAV-based yield prediction research for quinoa contrasts sharply with extensive work on architecturally comparable crops. Studies on soybean, a crop with similarly complex canopy architecture, have demonstrated that combining 3D structural features from point cloud data with multispectral information significantly improves yield prediction accuracy [27,28]. Similarly, cotton yield prediction studies have employed convolutional neural networks with UAV-based RGB and multispectral imagery, demonstrating the effectiveness of machine learning approaches for crops with indeterminate growth patterns [29]. These parallels in crops with complex, variable canopy architectures suggest considerable potential for UAV-ML applications in quinoa yield prediction, yet the lack of empirical validation represents a critical knowledge gap.

Despite extensive application of UAV-based remote sensing for yield prediction of major cereal crops [30–32], and emerging applications for quinoa phenotyping and stress assessment, quinoa yield prediction research remains limited to process-based modeling approaches using the Saltmed and FAO AquaCrop frameworks [33–35], and conventional field studies examining agronomic management effects [36–38]. These methodologies, while valuable for understanding water-yield relationships, rely on meteorological and soil inputs to simulate crop growth at field scales, limiting their capacity for spatially explicit, in-season yield forecasting. In contrast, data-driven UAV-based machine learning approaches enable plant-level predictions using high-resolution imagery captured throughout the growing season, offering opportunities for spatially explicit, in-season yield forecasting. To date, no studies have investigated the potential of UAV-based remote sensing approaches combined with ML for quinoa yield prediction, representing a significant gap in precision agriculture applications for this increasingly important crop.

To address this critical gap, this study investigates the feasibility of using UAV-based remote sensing and ML for plant-level quinoa yield prediction across multiple growth stages. The primary objectives are to:

1. Evaluate the performance of various ML regression models (Ridge Regression, K-Nearest Neighbors, Random Forest, XGBoost, and LightGBM) for yield prediction at each growth stage.
2. Identify the optimal growth stage(s) for quinoa yield prediction based on model performance metrics.
3. Explore the stage-specific importance of UAV-derived features to understand the underlying predictive mechanisms, particularly how structural and spectral features relate to quinoa's complex architecture.

By addressing these objectives, this research aims to determine the feasibility of UAV-based ML approaches for quinoa yield prediction and provide insights into optimal timing and key features for accurate prediction, thereby advancing precision agriculture techniques for this important crop.

2. Materials and methods

2.1. Experimental site and design

Field experiments were conducted at Tokyo University of Agriculture and Technology (TUAT), Fuchu, Tokyo, Japan (35°41'44" N, 139°29'32" E, 62 m elevation), during two growing seasons: 2022 and 2024–2025 (Fig. 1). Fuchu City has an average annual temperature of 14.0 °C, average annual rainfall of approximately 1647 to 1706 mm, and

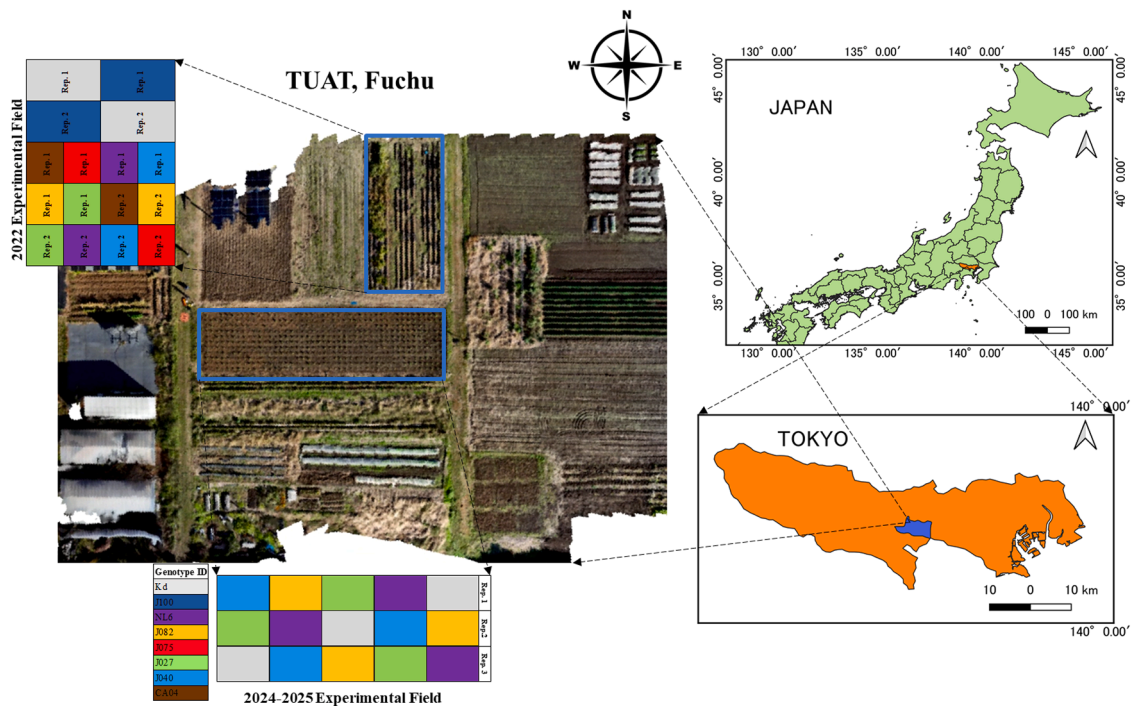


Fig. 1. Study area map and experimental field designs.

humidity of 63.7 %. The experimental site has predominantly silt loam soil.

Both experiments employed a randomized complete block design (RCBD) under rainfed conditions without fertilizer to assess the inherent performance of varieties under natural soil fertility. Seeds were obtained from the Japan International Research Center for Agricultural Sciences (JIRCAS). A plant spacing of 0.8 m x 0.8 m was maintained in both experiments, and border rows surrounded each plot to minimize edge effects.

In the 2022 experiment, eight quinoa varieties were evaluated with two replications. Plot sizes varied between 5.6 m x 4.8 m (42 plants) and 4.0 m x 2.4 m (15 plants) within a total experimental area of 23.2 m x 9.6 m. Seedlings were raised in a phytotron beginning August 23, with controlled conditions of 30 °C daytime temperature, 25 °C nighttime temperature, and 50 % relative humidity. Seedlings were transplanted to the field on September 15, three weeks after sowing. Harvesting occurred on December 30 at physiological maturity.

The 2024–2025 experiment evaluated five varieties selected from the 2022 study with three replications. Uniform plot sizes of 7.2 m x 3.2 m (36 plants) were used within a total area of 36.0 m x 9.6 m. Nursery establishment commenced on September 18 in the phytotron under modified conditions: 25 °C daytime temperature, 20 °C nighttime temperature, and 50 % relative humidity. Transplanting was conducted on October 8, three weeks after sowing. Harvesting was completed on January 8, 2025, at physiological maturity. Table 1 presents the list of genotypes used in both experiments.

Table 1
List of genotypes used.

Genotype	2022	2024–2025	Ecotype	Batch
Kd	✓	✓	Lowland (L)	Kd-S8d
J100	✓	×	Southern Highland (SH)	100-S9a
NL6	✓	✓	Lowland (L)	NL-6 I-S7a
J082	✓	✓	Lowland (L)	82-S7a
J075	✓	×	Northern Highland (NH)	75-S7a
J027	✓	✓	Intermediate (L/SH)	27-S5b
J040	✓	✓	Intermediate (L/SH)	40-S6c
CA04	✓	×	Lowland (L)	CA04-S4e

2.2. UAV data collection

Time-series remote sensing data were acquired using unmanned aerial vehicles (UAVs) equipped with different sensor configurations for each experimental season. Data collection was scheduled to correspond with five key growth stages (early vegetative, late vegetative, flowering, grain filling, and maturity), with flights conducted between 9:00 AM and 10:00 AM under optimal weather conditions to ensure consistent illumination and minimize shadow effects.

The 2022 Season campaign employed two complementary imaging systems. Multispectral data were captured using the MicaSense Altum sensor (MicaSense, Inc., Seattle, WA, USA; an AgEagle Aerial Systems subsidiary) mounted on a UAV platform. The Altum sensor simultaneously captures five multispectral bands (blue: 475 nm, green: 560 nm, red: 668 nm, red edge: 717 nm, near-infrared: 842 nm) and thermal infrared imagery (FLIR LWIR thermal camera: 8–14 μm) with integrated radiometric calibration capabilities.

Additionally, high-resolution RGB imagery was acquired using a DJI Zenmuse X5S camera (Model FC6520; SZ DJI Technology Co., Ltd., Shenzhen, China) with a 20.8 MP Micro Four Thirds sensor. For a detailed 3D reconstruction of individual quinoa plants, oblique imagery was captured using 360-degree circular flight paths around the experimental field.

Flight parameters were set at 15 m above ground level (AGL) with 80 % forward and side overlap. Five ground control points (GCPs) were distributed throughout the experimental area for precise georeferencing.

The 2024–2025 Season campaign utilized an integrated DJI Mavic 3 M platform (SZ DJI Technology Co., Ltd., Shenzhen, China), which combines a 4/3 CMOS 20 MP RGB camera with a 5 MP four-band multispectral imaging system (green: 560 nm, red: 650 nm, red edge: 730 nm, near-infrared: 860 nm). The system includes an RTK module for centimeter-level positioning accuracy and an integrated sunlight sensor for real-time irradiance measurements, ensuring radiometric consistency across flights.

Flight parameters were adjusted to 20 m AGL with 80 % forward and side overlap, with the camera oriented at 70 degrees from nadir to optimize plant structure reconstruction. Six GCPs were established using

a DJI D-RTK 2 high-precision GNSS mobile station and strategically distributed across the experimental area.

Each season comprised six total UAV flights: one pre-transplanting flight for terrain modeling and five subsequent flights aligned with critical growth stages (early vegetative, late vegetative, flowering, grain filling, and maturity). This temporal resolution enabled comprehensive monitoring of plant development and phenological progression throughout the experimental periods. Table 2 presents a summary of UAV flight campaign information.

2.3. Ground truth data collection

Individual plant yield measurements were obtained at physiological maturity to establish ground truth datasets for model training and validation. Plant selection followed systematic sampling protocols within each plot, avoiding border effects and selecting representative plants based on uniform distribution criteria.

In 2022, yield data were collected from 80 individual quinoa plants (5 plants x 2 replications x 8 varieties), while the 2024–2025 season provided 300 plant samples (20 plants x 3 replications x 5 varieties). Selected plants were tagged and georeferenced for precise alignment with UAV imagery.

Plants were harvested individually at maturity when the moisture content reached <15 %. Following harvest, samples were dried at 60 °C for 72 h, then threshed, cleaned, and weighed to determine final grain yield per plant using a precision balance. Individual plant grain yields were subsequently converted to g/m² for standardized reporting.

2.4. UAV image processing and feature extraction

UAV imagery processing was conducted using Agisoft Metashape Professional (Agisoft LLC, St. Petersburg, Russia) version 2.1.3, following standardized photogrammetric workflows. Initial quality assessment removed blurred or poorly exposed images from the dataset. Radiometric calibration was performed using calibration panels, while geometric corrections were applied using established ground control points (GCPs). The processing workflow comprised the following stages: image alignment with camera optimization, 3D point cloud generation, 3D mesh construction, and texture mapping. This workflow produced high-resolution orthomosaic images and digital elevation models (DEMs). All datasets were georeferenced to the JGD2011/Japan Plane Rectangular CS IX (EPSG:6677) projection system.

RGB-derived DEMs from the DJI Zenmuse X5S (2022) and DJI Mavic 3 M (2024–2025) sensors enabled the extraction of **three-dimensional structural parameters**, including plant height (PH), surface area (PSA), and volume (PV). Plant height (PH) was calculated by subtracting the bare ground DEM (from pre-transplanting flights) from vegetation-inclusive DEMs using QGIS version 3.34.4. PSA and PV measurements

for sampled plants were obtained using Agisoft Metashape's triangulated mesh method with a minimum component size of 100 faces. To ensure accurate measurements, the developed mesh required a defined coordinate system and closed geometry. Individual plants were carefully digitized using polygons, and the software's measurement tools calculated corresponding surface area and volume values.

Multispectral imagery from both seasons was radiometrically calibrated to surface reflectance using calibrated reflectance panels captured during each flight. The MicaSense Altum (2022) and DJI Mavic 3 M (2024–2025) both incorporate integrated light sensors (DLS2 and sunlight sensor, respectively) that record incident solar irradiance, enabling atmospheric compensation during calibration. Agisoft Metashape processed calibration panel images to derive sensor-specific coefficients, converting raw sensor values to surface reflectance. The software exports reflectance data as 16-bit scaled integers (0–65,535 range), where 65,535 represents 100 % reflectance. Before vegetation index calculation, these values were converted to a conventional 0–1 reflectance scale by dividing by 65,535 (e.g., NIR value 8952.721 → 0.1366 reflectance). To maintain consistency across seasons, only the four spectral bands common to both sensors (green, red, red edge, and NIR) were used for vegetation index calculations, with the MicaSense Altum's blue and thermal bands excluded from analysis. Spectral reflectance values were extracted from calibrated multispectral orthomosaics using 0.8 m × 0.8 m polygon grids centered on each of the 380 sampled plants. QGIS zonal statistics tools extracted mean, standard deviation, minimum, and maximum reflectance values for the four spectral bands within each polygon. The same approach extracted plant height statistics from DEM layers. Following reflectance extraction and conversion to a 0–1 scale, 13 vegetation indices were calculated using standardized formulas (Table 3). All vegetation indices were derived from multispectral sensors only (MicaSense Altum in 2022; DJI Mavic 3 M multispectral camera in 2024–2025), while 3D structural parameters were derived from RGB sensors only (DJI Zenmuse X5S in 2022; DJI Mavic 3 M RGB camera in 2024–2025). All measurements were systematically recorded and organized in Excel spreadsheets for subsequent analysis.

2.5. Yield prediction framework and modeling approach

2.5.1. Feature set and data preparation

The complete feature set for yield prediction modeling comprised 16 variables: 13 vegetation indices derived from multispectral imagery and three structural parameters (PH: plant height, PSA: plant surface area, and PV: plant volume) extracted from RGB imagery through photogrammetric reconstruction. The extracted features and corresponding yield data were organized into a dataset. The dataset was then filtered to create separate subsets for each of the five defined growth stages in both seasons.

Table 2
UAV flight campaign information.

Year	Growth Stage	UAV Flight Height (m)	Forward/Side Overlap (%)	UAV Flight Path Orientation	Number of Aligned Images	
					RGB	Multi-spectral
2022	Early vegetative	15	80	360° circular path	237	215
	Late vegetative	15	80	360° circular path	240	215
	Flowering	15	80	360° circular path	240	220
	Grain filling	15	80	360° circular path	240	245
	Maturity	15	80	360° circular path	240	220
2024–2025	Early vegetative	20	80	Camera oriented @ 70°	1197	1115
	Late vegetative	20	80	Camera oriented @ 70°	234	312
	Flowering	20	80	Camera oriented @ 70°	243	324
	Grain filling	20	80	Camera oriented @ 70°	249	332
	Maturity	20	80	Camera oriented @ 70°	249	332
					291	388
					1266	1688
2022 & 2024–2025					2463	2803

Table 3
List of vegetation indices used.

Vegetation index	Formula	Reference
Difference vegetation index (DVI)	NIR-R	[39]
Ratio vegetation index (RVI)	NIR/R	[40]
Renormalized difference vegetation index (RDVI)	(NIR - R) / SQT(NIR+R)	[41]
Modified soil-adjusted vegetation index (MSAVI)	$(2 \times \text{NIR} + 1 - \sqrt{(2 \times \text{NIR} + 1)^2 - 8 \times (\text{NIR} - \text{R})}) / 2$	[42]
Modified soil-adjusted vegetation index 2 (MSAVI 2)	$(2 \times \text{NIR} + 1 - \sqrt{(2 \times \text{NIR} + 1)^2 - 8 \times (\text{NIR} - \text{R})}) / 2$	[42]
Optimized soil adjusted vegetation index (OSAVI)	$(\text{NIR} - \text{R}) / (\text{NIR} - \text{R} + 0.16)$	[43]
Enhanced vegetation index (EVI)	$2.5 \times (\text{NIR} - \text{R}) / (1 + \text{NIR} - 2.4 \times \text{R})$	[44]
Normalized difference vegetation index (NDVI)	$(\text{NIR} - \text{R}) / (\text{NIR} + \text{R})$	[45]
Soil adjusted vegetation index (SAVI)	$1.5 (\text{NIR} - \text{RE}) / (\text{NIR} + \text{RE} + 0.5)$	[46]
Soil-adjusted red-edge index (SARE)	$(\text{NIR}-\text{RE})/(\text{NIR} + \text{RE} + 0.25) + 0.25$	[47]
NIR-RE normalized difference vegetation index (NIRRENDVI)	$((\text{NIR} + \text{RE})/2 - \text{R}) / ((\text{NIR} + \text{RE})/2 + \text{R})$	[48]
Red-edge normalized difference vegetation index (RENDVI)	$(\text{RE}-\text{R})/(\text{RE}+\text{R})$	[49]
Red-edge ration vegetation index (RERVI)	NIR/RE	[50]

To quantify the contribution of experimental design factors to yield variability, two-way ANOVA (Variety $\times$ Replication) was performed separately for each season due to the unbalanced design across years. Plants were arranged in a randomized complete block design (RCBD) with 2 replications in the 2022 season and 3 replications in the 2024–2025 season. Eight quinoa varieties (Table 1) were evaluated in 2022 ($n = 80$ plants), while five varieties (a subset of the 2022 set) were evaluated in 2024–2025 ($n = 300$ plants). The design was balanced within each season with equal sample sizes per variety-replication combination.

Pearson correlation analysis was performed to assess the linear relationship between each of the 16 UAV-derived features and yield for each of the five growth stages independently.

2.5.2. Machine learning models development and evaluation

Individual yield prediction models were developed for each quinoa growth stage across both seasons to identify optimal timing windows for yield prediction. Five traditional machine learning algorithms were employed, each treating temporal observations as independent data points:

- Ridge Regression (RR):** A linear regression model incorporating L2 regularization to prevent overfitting by penalizing large coefficients [51]. This approach assumes linear relationships between features and yield while controlling model complexity.
- k-Nearest Neighbors (kNN):** A non-parametric algorithm that predicts yield values based on the k nearest neighbors in the feature space [52]. This method captures local patterns without assuming global functional relationships.
- Random Forest (RF):** An ensemble method that constructs multiple decision trees during training and averages their predictions to reduce overfitting and improve generalization [53].
- Extreme Gradient Boosting (XGBoost):** An optimized gradient boosting algorithm that has demonstrated superior predictive accuracy for forecasting crop yields when trained on agricultural parameters including pesticides, rainfall, and average temperature [54].
- Light Gradient Boosting Machine (LightGBM):** A lightweight gradient boosting machine algorithm that grows trees leaf-wise and requires fewer computational resources, making it more efficient for large-scale yield prediction tasks [55].

The complete dataset comprised 380 individual plants (80 from the 2022 season, 300 from the 2024–2025 season) representing eight quinoa varieties (CA04, J027, J040, J075, J082, J100, Kd, NL6). For each growth stage, data from both seasons were pooled to create stage-specific datasets, enabling the identification of optimal phenological timing for yield prediction when multi-year training data are available.

Model performance was evaluated using a nested cross-validation framework designed to provide unbiased performance estimates and maintain independence of the test set. For each growth stage, a nested 5-fold cross-validation strategy was implemented to ensure robust model evaluation:

- Outer loop (model evaluation):** Data were partitioned into five folds using stratified 5-fold cross-validation (`random_state = 42`). Stratification was performed based on plant labels to ensure proportional representation of all eight varieties in both seasons in each fold. Each fold served sequentially as an independent test set (20 % of data, ~ 76 plants) while the remaining four folds (80 %, ~ 304 plants) constituted the training set.
- Inner loop (hyperparameter tuning):** Within each training set of the outer loop, 3-fold cross-validation (`random_state = 43`) was performed exclusively on the training data to identify optimal hyperparameters through grid search, using R^2 as the optimization metric. The hyperparameter search spaces were defined as follows: Ridge Regression explored 20 logarithmically-spaced alpha values from 10^{-4} to 10^0 ; K-Nearest Neighbors tested `n_neighbors` from 1 to 20; Random Forest evaluated combinations of `n_estimators` (50, 100, 200), `max_depth` (None, 10, 20), and `min_samples_split` (2, 5, 10); XGBoost examined `n_estimators` (100, 200, 300), `learning_rate` (0.01, 0.1, 0.2), and `max_depth` (3, 5, 7); and LightGBM tested `n_estimators` (100, 200, 300), `learning_rate` (0.01, 0.1, 0.2), and `num_leaves` (30, 50, 100). The test fold from the outer loop was never accessed during this tuning process, preventing information leakage between hyperparameter selection and model evaluation.
- Model fitting and evaluation:** After hyperparameter optimization, each model was retrained on the full training set in the inner loop (4 folds) with optimal parameters and evaluated on its corresponding held-out test fold. This process was repeated for all five outer folds.

Model performance was evaluated on both the training and testing datasets using three widely accepted regression metrics:

$$R^2 = 1 - \left\{ \sum_{i=1}^n (y_i - \hat{y}_i)^2 \right\} \div \left\{ \sum_{i=1}^n (y_i - \bar{y})^2 \right\} \quad (1)$$

$$RMSE = \sqrt{\left\{ \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \right\}} \quad (2)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n (|y_i - \hat{y}_i|) \quad (3)$$

Where: n = number of observations; y_i and $\hat{y}_i$ denote the i th actual values and predicted quinoa yield, respectively; and $\bar{y}$ = mean of actual yield. For each growth stage and algorithm, these metrics were calculated separately for each of the five outer folds, and final reported metrics represent the mean values across these five folds, providing robust estimates of model generalization performance.

2.5.3. Feature importance analysis

To understand the contribution and importance of each UAV-derived feature to the yield predictions, two complementary feature importance analyses were conducted for the best-performing models at each growth stage. The best-performing model for each stage was identified based on the lowest average test RMSE value.

First, SHapley Additive exPlanations (SHAP) [56] analysis was

performed to quantify the impact of each feature on model predictions. For each growth stage, SHAP values were calculated for the predictions of the best-performing model on the independent test dataset, using the training data as the background dataset for the SHAP explainer. Mean absolute SHAP values were calculated to represent the overall importance of each feature within each stage's best model, providing insight into how features contribute to individual predictions.

Second, permutation importance analysis was conducted to assess the importance of features based on model performance degradation. For each growth stage, permutation importance was calculated separately for each of the five cross-validation folds using the corresponding best-performing model and its held-out test set. The analysis was performed with 10 random permutations per feature (random_state = 42), using R² as the scoring metric. For each fold, the mean importance across the 10 permutations was recorded, and final feature importance values for each stage represent the average across all five folds. This approach quantifies how much the model's predictive performance decreases when a feature's values are randomly shuffled, thereby breaking the relationship between that feature and the target variable.

2.5.4. Software and implementation

All analyses were implemented using Python 3.12.12 in Google Colaboratory (November 2024 - January 2025). Machine learning models were built using scikit-learn 1.6.1 (Ridge Regression, K-Nearest Neighbors, Random Forest), XGBoost 3.1.2, and LightGBM 4.6.0. Data manipulation was performed with NumPy 2.0.2 and pandas 2.2.2, while visualizations were created using Matplotlib 3.10.0 and Seaborn 0.13.2. Feature importance analysis employed SHAP 0.50.0, with TreeExplainer for tree-based models and KernelExplainer for other models. The complete analysis code and implementation details are publicly available at <https://github.com/Gucchyon/QuinoaYieldPrediction>.

3. Results

3.1. Analysis of variance

Analysis of variance (ANOVA) for quinoa grain yield across growing seasons is presented in Table 4. Results showed that variety effects dominate yield variability. Genotypic differences explained 68–83 % of total yield variance across seasons, confirming substantial genetic differences in yield potential among the eight quinoa varieties evaluated. Block/replication effects were negligible in 2022 (6 %, p = 0.469) and small in 2024–2025 (31 %, though statistically significant due to large sample size), indicating relatively uniform field conditions. The significant replication effect in 2024–2025 likely reflects minor environmental gradients across the larger experimental area but represents a small absolute contribution to total variance. Unexplained variance (11 % in 2022, 1 % in 2024–2025) represents within-variety, within-block variability driven by individual plant differences in micro-environmental exposure, planting quality, developmental timing, and other localized factors. This is precisely the variation our UAV-based ML

models aim to predict through high-resolution phenotyping.

3.2. Correlation analysis

Pearson correlation analysis revealed varying relationships between UAV-derived features and grain yield across growth stages (Fig. 2). The strongest correlations were observed during the flowering stage, with NDVI (r = 0.84), SARE (r = 0.84), and RERVI (r = 0.84) showing the highest associations with yield. Late vegetative and early vegetative stages also demonstrated strong correlations, particularly for Plant Surface Area (PSA; r = 0.86 and r = 0.85, respectively). RERVI (r = 0.83) and NDVI (r = 0.82) also showed strong associations with yield during the late vegetative stage, while Plant Volume (PV; r = 0.84) showed a strong correlation with yield in the early vegetative stage.

In contrast, correlations during grain filling and maturity stages were notably weaker. The highest correlations during grain filling were observed for PV (r = 0.78), PSA (r = 0.74), and Plant Height (PH; r = 0.73), while maturity showed the weakest overall correlations, with PSA (r = 0.77) and PV (r = 0.69) being the strongest predictors. This decline in correlation strength at maturity likely reflects the onset of senescence, where leaf yellowing, stem desiccation, and heterogeneous panicle coloration disrupt the relationships between canopy characteristics and final yield. Across all growth stages, 3D structural parameters (PSA, PV, PH) and traditional vegetation indices (RERVI, SARE, NDVI) consistently showed stronger correlations with yield.

3.3. Machine learning model performance and optimal prediction timing

The performance of five machine learning models (Ridge Regression, K-Nearest Neighbors, Random Forest, XGBoost, and LightGBM) for predicting yield across five growth stages was evaluated using a nested 5-fold cross-validation. Visualizations of the performance metrics show the comparative performance of the models across stages for both training and testing datasets (Fig. 3). Random Forest generally emerged as the best-performing model, achieving optimal test metrics during flowering stage (R² = 0.726, RMSE = 16.71 g/m², MAE = 10.91 g/m²), which represented the highest accuracy across all stages and models. Random Forest also performed best during late vegetative (R² = 0.704, RMSE = 17.22 g/m², MAE = 11.81 g/m²), grain filling (R² = 0.698, RMSE = 17.52 g/m², MAE = 9.75 g/m²), and maturity (R² = 0.658, RMSE = 19.29 g/m², MAE = 13.04 g/m²) stages. Ridge Regression achieved the best performance during early vegetative (R² = 0.690, RMSE = 17.89 g/m², MAE = 13.13 g/m²), providing comparable accuracy to the late vegetative and grain filling stages predictions.

Predictive accuracy progressively improved from early vegetative through late vegetative to the flowering stage, then declined during maturity. This temporal pattern identifies flowering and grain filling as optimal windows for yield prediction, offering both high accuracy and sufficient lead time before harvest for management decisions. Scatter plots of actual versus predicted yield for the best model in each stage demonstrate strong predictive accuracy, with observations closely

Table 4
Analysis of variance (ANOVA) for quinoa grain yield (g/m²) across growing seasons.

Source of Variation	DF	Sum of Square	Mean Square	F Value	Pr(> F)	Variance Explained (%)
2022 (N=80)						
Variety	7	66,898.81	9556.972	7.46	<0.001**	83
Replication (Block)	1	679.712	679.712	0.53	0.4687	6
Residual	71	90,929.37	1280.695			11
Total	79	158,507.9				100
2024–2025 (N=300)						
Variety	4	21,893.2	5473.300	53.65	<0.001**	68
Replication	2	4975.077	2487.538	24.38	<0.001**	31
Residual	293	29,890.78	102.0163			1
Total	299	56,759.06				100

**p < 0.001

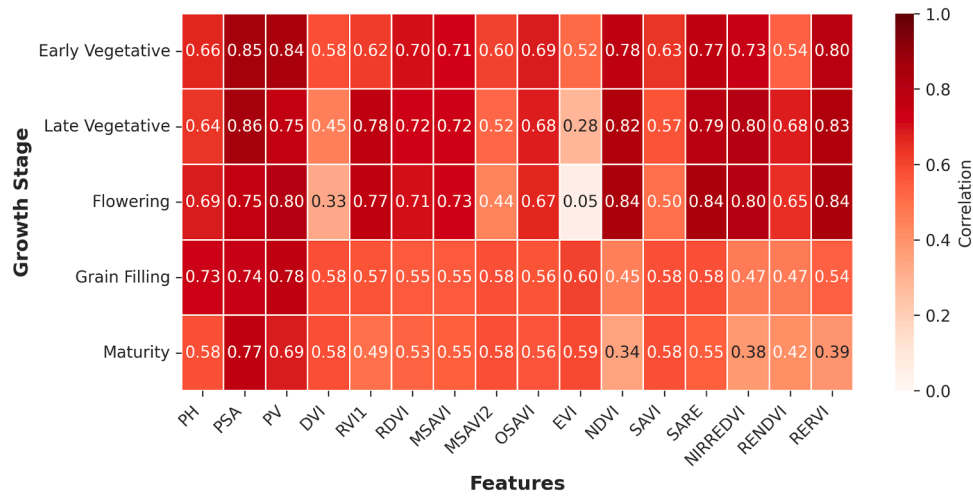


Fig. 2. Correlation coefficients between UAV-derived features and quinoa grain yield across five growth stages. Values represent Pearson correlation coefficients, with warmer colors indicating stronger positive correlations.

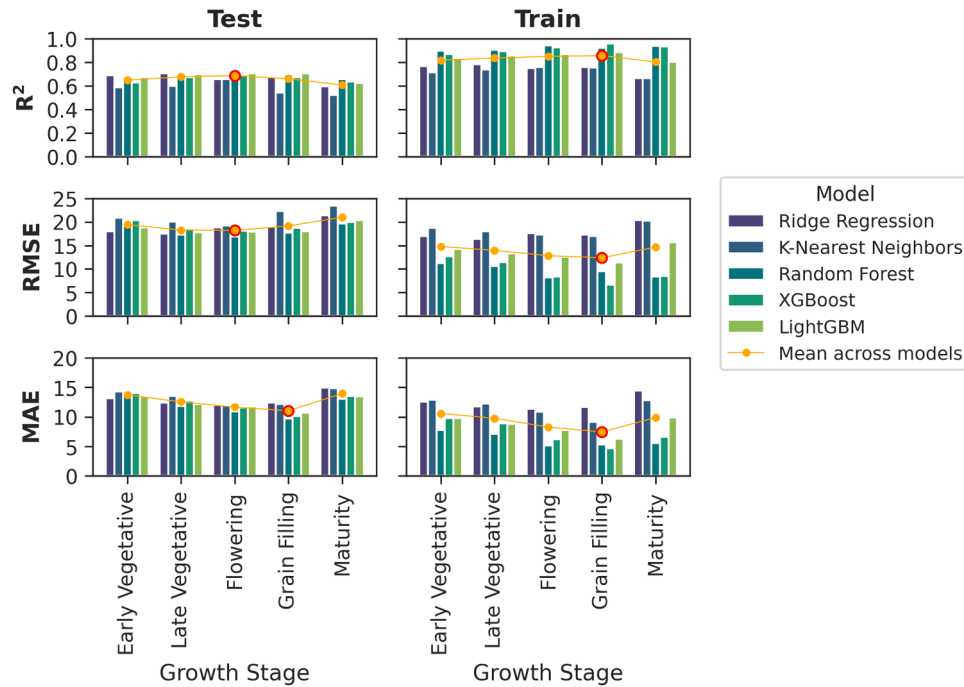


Fig. 3. Performance comparison of five machine learning algorithms for quinoa yield prediction across growth stages. Bar plots show individual model performance, while orange lines with circles represent the mean performance across all models. The best-performing stage for each metric is highlighted with a red outline. The left column shows test set performance, while the right column shows training set performance.

aligned to the 1:1 line, particularly during the flowering stage with higher R^2 and lower error values (Fig. 4).

3.4. Feature importance analysis

SHAP (SHapley Additive exPlanations) analysis quantified the contribution of individual UAV-derived features to yield predictions across growth stages, revealing distinct phenological patterns in feature importance (Fig. 5). During the early vegetative stage, MSAVI was the dominant predictor (SHAP = 15.88), followed by RERVI (14.69), NDVI (12.59), and PSA (11.14). At the late vegetative stage, PSA exhibited the highest importance (16.63), with contributions from SARE (2.56), EVI (1.89), PV (1.68), and RERVI (1.66). Flowering stage showed PV as the primary predictor (14.52), accompanied by RERVI (2.74), RVI (2.56), PSA (2.20), NDVI (2.14), and SARE (1.87). During grain filling, PV

dominated feature importance (16.23), followed by RERVI (3.80), PH (3.78), and PSA (1.86). At maturity, PSA (11.35) and PV (10.87) were equally important, with moderate contribution from PH (1.76).

Permutation importance analysis, which measures model performance degradation when feature-target relationships are disrupted, largely corroborated the SHAP-based findings while providing additional insights into feature robustness. At the early vegetative stage, MSAVI (1.53) and NDVI (1.18) demonstrated the strongest influence on model performance, followed by RVI (0.43), PSA (0.40), and RERVI (0.76). During late vegetative growth, PSA remained the most critical feature (0.83), with SARE (0.004) and EVI (0.02) showing modest but positive contributions. At flowering, PV emerged as the dominant feature (0.72), while PSA (0.01) and RERVI (0.02) contributed minimally. The grain filling stage exhibited strong dependence on PV (0.83), with secondary contributions from PH (0.09) and RERVI (0.06). At

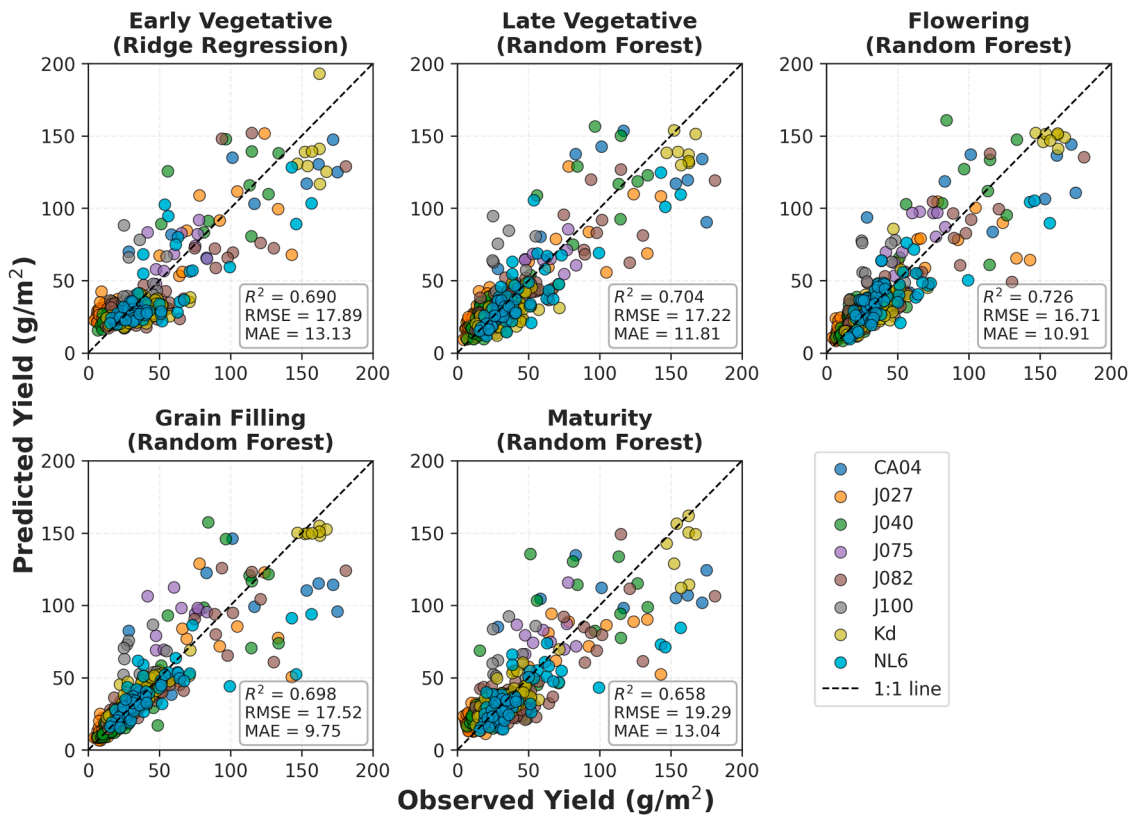


Fig. 4. Scatter plots of observed versus predicted quinoa yield across five growth stages. Each panel displays predictions from the best-performing model (indicated in parentheses) obtained through 5-fold cross-validation. Data points are colored by variety. The black dashed line represents the 1:1 line. Performance metrics (R^2 , RMSE, and MAE) represent the average across all folds.

maturity, both PV (0.40) and PSA (0.36) were critical, followed by PH (0.05).

The convergence of SHAP and permutation importance results across growth stages strengthens confidence in the identified feature-importance patterns. Notably, 3D structural parameters (PH, PSA, PV) were consistently among the most important features across multiple growth stages, generally showing a positive correlation with predicted yield. Meanwhile, spectral indices showed stage-specific importance primarily during vegetative and reproductive phases. This pattern demonstrates that volumetric and surface measurements capture yield-relevant information more comprehensively than traditional two-dimensional spectral approaches.

4. Discussion

4.1. Comparative performance of machine learning algorithms

Ensemble methods, specifically Random Forest, demonstrated superior predictive performance compared to linear models (Ridge Regression) and instance-based algorithms (KNN) (Fig. 3, 4). This result is consistent with previous studies documenting the effectiveness of tree-based ensemble methods for modeling complex, non-linear relationships in agricultural remote sensing data [57,58]. The enhanced performance of ensemble methods suggests that relationships between UAV-derived features and yield are characterized by non-linear interactions and threshold effects that cannot be adequately captured by linear modeling approaches.

Evaluation of training versus testing performance revealed evidence of overfitting in some models (particularly XGBoost), with training R^2 values substantially exceeding testing R^2 values. This discrepancy indicates that these models may be fitting to noise in the training data rather than capturing generalizable patterns [22]. Future

implementations should employ expanded training datasets from multiple growing seasons or ensemble averaging approaches to mitigate overfitting. However, the magnitude of overfitting decreased during the flowering stage, with training-testing performance gaps narrowing. This reduction supports the hypothesis that phenologically optimized prediction windows improve model generalization by ensuring that measurements are taken when signal-to-noise ratios in the relationship between predictors and yield are maximized [59,60].

Ridge Regression and KNN exhibited minimal overfitting but achieved lower overall accuracy during later growth stages compared to ensemble methods. This trade-off between model complexity and generalization capacity highlights the importance of validation strategies that assess both predictive accuracy and model stability across independent datasets. While ensemble methods require greater computational resources, modern UAV data processing workflows can readily accommodate these algorithms, particularly when processing is conducted post-flight rather than in real-time.

4.2. Temporal variation in yield prediction accuracy

Yield prediction accuracy demonstrated marked variation across quinoa development stages, with optimal predictive performance achieved during reproductive phases (Fig. 3, 4). The flowering stage yielded the strongest predictions ($R^2 = 0.726$, MAE = 10.91 g/m²), followed closely by grain filling ($R^2 = 0.698$, MAE = 9.75 g/m²). The ANOVA analysis confirms that while variety effects dominate yield variability (as expected from genetic differences), substantial within-variety variance remains that UAV-based machine learning approaches can successfully model. The strong machine learning performance (R^2 up to 0.726) indicates that high-resolution remote sensing captures morphological and spectral features that integrate both genetic potential and environmental responses, enabling accurate plant-level yield prediction. These

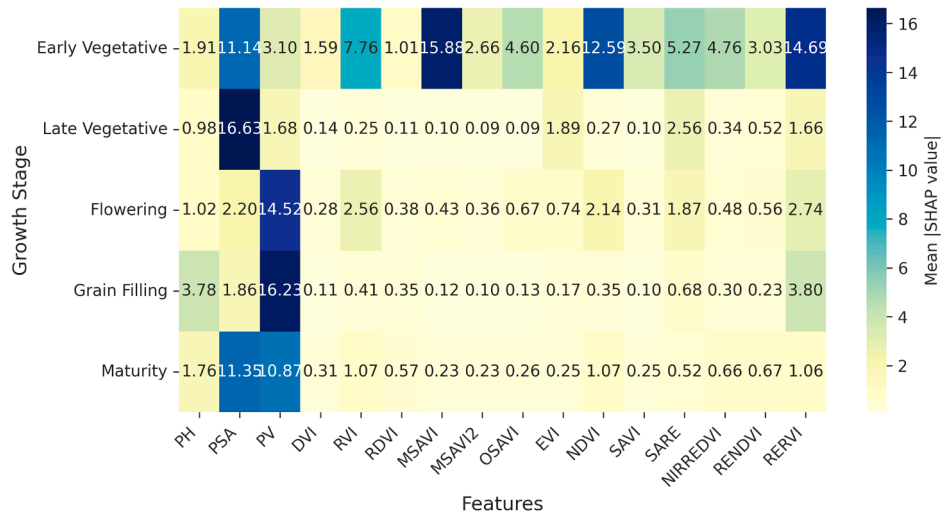
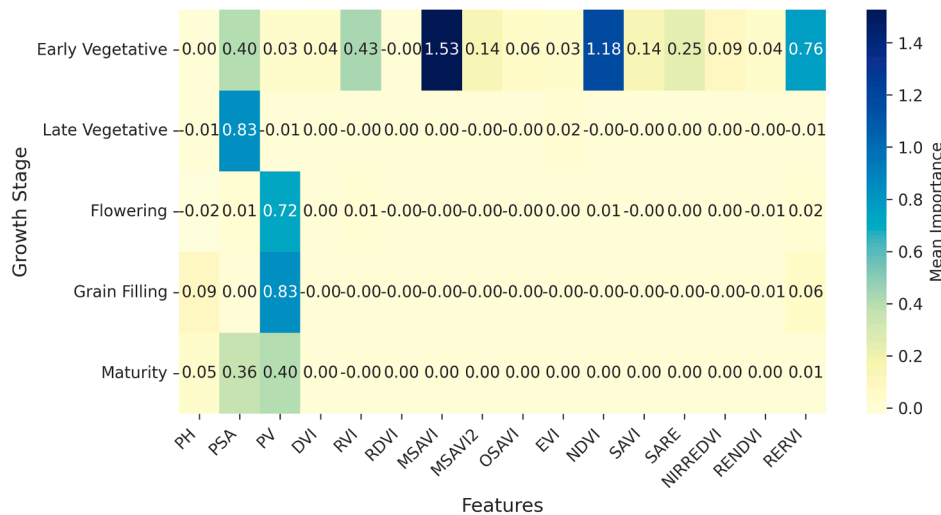
(a) SHAP**(b) Permutation Importance**

Fig. 5. Feature importance analysis across growth stages. (a) Mean absolute SHAP values and (b) mean permutation importance values for UAV-derived features from the best-performing model at each growth stage, averaged across 5-fold cross-validation. Higher values (darker colors) indicate greater feature importance for yield prediction.

accuracy levels are comparable to those reported for UAV-based yield prediction in major cereals, where R^2 values typically range from 0.73 to 0.75 for rice and wheat [61,30], demonstrating that quinoa's complex architecture does not inherently limit prediction accuracy when appropriate features are employed.

This temporal pattern aligns with established principles in cereal crop phenology, where reproductive stages represent optimal windows for yield prediction [61–63]. The progressive improvement in prediction accuracy during later developmental stages can be attributed to the strengthening correlation between UAV-derived metrics and final grain yield as canopy architecture stabilizes and biomass accumulation approaches its maximum.

The superior predictive performance during the flowering stage corresponds with the critical yield determination period identified by Bertero and Ruiz [64], who demonstrated that quinoa yield is predominantly determined during the flowering phase, from first anthesis to the end of anthesis. The high R^2 values obtained at this stage indicate that UAV-derived structural and spectral features effectively quantify canopy characteristics that are directly linked to reproductive development and subsequent grain production. Similarly, the comparable high prediction accuracy during grain filling aligns with the sensitive period reported by

Vásquez et al. [65], occurring between -26 and 445 °Cd from flowering. During this phase, final seed weight is determined through carbohydrate allocation and accumulation processes, which are reflected in measurable canopy characteristics captured and translated into yield predictions by UAV sensors.

However, prediction accuracy decreases during maturity as physiological senescence progresses. This decline can be attributed to several factors: (1) leaf yellowing and abscission reduce the photosynthetically active canopy area, (2) stem desiccation alters structural relationships between canopy volume and biomass, and (3) heterogeneous panicle coloration, ranging from green to yellow, red, and brown depending on genotype and environmental conditions-disrupts spectral signatures that previously correlated with yield potential. The variable onset and progression of senescence across individual plants within plots further increases measurement noise, weakening the predictive signal.

The identified temporal pattern of prediction accuracy creates distinct opportunities for agricultural decision support. The fact that comparable yield prediction performance was achieved during early and late vegetative stages ($R^2 = 0.690$ and 0.704) suggests that early-season measurements of canopy architecture and spectral properties capture important baseline information about crop establishment, vigor, and

initial growth conditions that persist to influence final yield. It demonstrates that UAV monitoring can provide useful early-season management insights. Flowering stage assessment enables timely interventions such as targeted nutrient application or irrigation adjustments to optimize seed set during the critical yield determination period. Grain filling stage predictions provide accurate yield projections well before harvest maturity, supporting marketing decisions and harvest logistics planning. This flexibility is particularly valuable for quinoa production systems, where the crop's inherently prolonged and variable development patterns require adaptable management approaches that can respond to emerging crop conditions throughout the growing season.

4.3. Relative importance of UAV-Derived features

SHAP analysis identified plant volume (PV) as the most important predictor variable across multiple growth stages, with maximum importance values observed during flowering (SHAP = 14.52) and grain filling (SHAP = 16.23) (Fig. 5). Permutation importance analysis corroborated these findings, demonstrating that PV had the strongest impact on model performance at flowering (0.72) and grain filling (0.83), and remained critical at maturity (0.40). PV, derived from structure-from-motion photogrammetry of RGB imagery, provides a three-dimensional measure of canopy architecture that integrates both vertical and lateral growth components [66]. The convergence of SHAP and permutation importance results that volumetric measures of canopy structure contain more yield-relevant information than two-dimensional metrics, likely because they more accurately represent total above-ground biomass [67,68].

Plant surface area (PSA) exhibited elevated importance during late vegetative growth (SHAP = 16.63; permutation importance = 0.83) and physiological maturity (SHAP = 11.35; permutation importance = 0.36). During vegetative development, PSA quantifies the total photosynthetically active surface area available for light interception and carbon assimilation [69], which establishes the foundation for subsequent grain filling. At maturity, PSA may serve as an indicator of canopy persistence and delayed senescence, characteristics associated with enhanced stress tolerance and yield stability across diverse environmental conditions [70,71].

Plant height (PH) demonstrated moderate importance during early vegetative (SHAP = 1.91), grain filling stages (SHAP = 3.78; permutation importance = 0.09), and maturity (SHAP = 1.76; permutation importance = 0.05), but was consistently outperformed by volumetric metrics (PV and PSA). This pattern suggests that simple height measurements capture only a fraction of the architectural complexity relevant to yield determination in quinoa [20]. The superior performance of three-dimensional metrics indicates that canopy volume and surface area provide more comprehensive characterization of light interception capacity, spatial resource acquisition, and biomass distribution-factors that collectively determine photosynthetic potential and yield [72–75]. The consistent importance of PH, PSA, and PV across growth stages underscores their value as core phenotypic traits for yield prediction. Given their demonstrated predictive power, future research should prioritize improving the precision and reliability of these measurements through refined sensing protocols and quality control procedures to maximize their utility in breeding programs and precision agriculture applications.

A critical finding of this study is that traditional vegetation indices (MSAVI, NDVI, RERVI) exhibited strong univariate correlations and the highest SHAP/permutation importance values during the early vegetative stage, yet showed markedly reduced importance in multivariate models during late-growth stages. This apparent discrepancy reveals that while these spectral indices covary with yield when considered independently, their predictive contribution becomes largely redundant when integrated with 3D structural parameters. The 3D metrics capture branching complexity and canopy density, which are directly linked to biomass accumulation and reproductive sink capacity in quinoa. This

finding demonstrates a fundamental advantage of structure-from-motion photogrammetry over spectral-only approaches: by reconstructing complete three-dimensional canopy architecture, these methods inherently encode spatial information that captures much of the same physiological variation (biomass, leaf area, vigor) that spectral indices attempt to quantify through reflectance patterns alone. The result is a more comprehensive phenotypic characterization with reduced feature redundancy and improved model interpretability.

The deterioration in predictive performance of vegetation indices during late-stage growth is likely due to heterogeneous coloration of ripening quinoa panicles disrupting spectral signatures. Unlike cereals such as wheat or rice, where senescence follows relatively uniform patterns, quinoa panicles display genotype-specific coloration ranging from bright yellow to deep red and purple hues. This spectral heterogeneity obscures the chlorophyll-based relationships underlying traditional vegetation indices. Consequently, this biological characteristic of quinoa further underscores the value of 3D structural approaches, which remain robust to color variation by relying on geometric properties that maintain stable relationships with yield throughout development.

4.4. Limitations and future research directions

While this study provides valuable insights into UAV-based yield prediction for quinoa, several avenues warrant further investigation to enhance model generalizability and practical application. First, the transferability of these models across diverse growing environments and quinoa cultivars should be evaluated, as phenological variation and cultivar-specific growth patterns may influence optimal prediction timing and feature importance. Second, integration of environmental variables (water availability, temperature, solar radiation) could improve predictions during grain filling, when yield becomes increasingly sensitive to environmental conditions that affect assimilate partitioning and grain development. Third, given the modest improvements from hyperparameter tuning in some growth stages and the relatively limited dataset size per stage, exploring ensemble methods or expanding training datasets across multiple seasons and locations would likely enhance model robustness and predictive accuracy. Finally, multi-year, multi-location validation studies are essential to confirm the identified prediction windows and strengthen agronomic recommendations for adoption by producers and agricultural advisors.

5. Conclusions

This study provides the first demonstration of UAV-based machine learning for plant-level quinoa yield prediction in precision agriculture. The flowering stage emerged as the optimal phenological period for yield prediction, achieving the highest accuracy ($R^2 = 0.726$) using Random Forest regression. The comparable yield prediction accuracy achieved during early and late vegetative stages ($R^2 = 0.690$ and 0.704 , respectively) demonstrates that UAV monitoring can provide useful early-season quinoa management insights.

Variable importance analysis revealed that 3D structural parameters, particularly plant volume and surface area, were the strongest predictors of quinoa yield, substantially outperforming traditional spectral vegetation indices in multivariate models. This demonstrates that structure-from-motion photogrammetry captures yield-relevant architectural information more comprehensively than spectral reflectance approaches, as volumetric metrics integrate spatial complexity that cannot be adequately represented by two-dimensional indices. This finding has important implications for operational remote sensing systems targeting crops with complex canopy architecture.

For operational implementation, we recommend UAV monitoring during the flowering stage, prioritizing the extraction of 3D structural metrics through structure-from-motion photogrammetry. These findings fill a critical knowledge gap in quinoa precision agriculture and provide producers with a practical, cost-effective tool for yield estimation weeks

before harvest, enabling improved planning and resource allocation decisions.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Gemini AI to assist with generating and debugging Python code for data analysis. The authors reviewed and edited all AI-generated code and take full responsibility for the accuracy and functionality of the final analysis scripts.

Funding

This research was supported by the Science and Technology Research Partnership for Sustainable Development (SATREPS); Japan Science and Technology Agency (JST); Japan International Cooperation Agency (JICA) (Grant No. JP-MJSA1907); Ministry of Agriculture, Forestry and Fisheries (MAFF)'s "Commissioned Projects for Promotion of Strategic International Joint Research" (Joint Research with Germany, Grant No. JPJ008837); and the grant from Yanmar Environmental Sustainability Support Association.

Data availability

The complete analysis code and implementation details are publicly available at <https://github.com/Gucchyon/QuinoaYieldPrediction>.

Data will be made available on request.

Ethical statement

This study did not involve human participants or animals (or the use of proprietary personal data) and therefore did not require ethics committee approval or written informed consent.

All experiments and data collection were carried out in accordance with institutional and national guidelines for agricultural/field research and in compliance with standard safety and data integrity practices.

CRediT authorship contribution statement

Sayo Sesay: Methodology, Conceptualization, Data curation, Visualization, Formal analysis, Writing – original draft. **Tomoaki Yamaguchi:** Writing – review & editing, Visualization, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Sayako Kushino:** Methodology, Data curation. **Yasuharu Yoshikawa:** Methodology, Data curation. **Shunsuke Adachi:** Supervision, Data curation. **Keisuke Katsura:** Writing – review & editing, Visualization, Supervision, Methodology, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors gratefully acknowledge the Japan International Research Center for Agricultural Sciences (JIRCAS) for providing the quinoa germplasm used in this study.

References

- [1] A. Vega-Gálvez, M. Miranda, J. Vergara, E. Uribe, L. Puente, E.A. Martínez, Nutrition facts and functional potential of quinoa (*Chenopodium quinoa* Willd.), an

- ancient Andean grain: a review, *J. Sci. Food Agric.* 90 (2010) 2541–2547, <https://doi.org/10.1002/jsfa.4158>.
- [2] L. Hinojosa, J.A. González, F.H. Barrios-Masias, F. Fuentes, K.M. Murphy, Quinoa abiotic stress responses: a review, *Plants* 7 (2018) 106, <https://doi.org/10.3390/plants7040106>.
- [3] FAO, FAOSTAT: Crops and Livestock Products, Food and Agriculture Organization of the United Nations, 2023 <https://www.fao.org/faostat/en/#data/QCL>.
- [4] S-E. Jacobsen, The situation for quinoa and its production in southern Bolivia: from economic success to environmental disaster, *J. Agron. Crop Sci.* 197 (2011) 390–399, <https://doi.org/10.1111/j.1439-037X.2011.00475.x>.
- [5] V. Angeli, P. Miguel Silva, D. Crispim Massuela, M.W. Khan, A. Hamar, F. Khajehei, S. Graeff-Hönninger, C. Piatti, Quinoa (*Chenopodium quinoa* Willd.): an overview of the potentials of the "golden grain" and socio-economic and environmental aspects of its cultivation and marketization, *Foods* 9 (2020) 216, <https://doi.org/10.3390/foods9020216>.
- [6] F. Furukawa, K. Maruyama, Y.K. Saito, M. Kaneko, Corn height estimation using UAV for yield prediction and crop monitoring, in: R. Avtar, T. Watanabe (Eds.), *Unmanned Aerial Vehicle: Applications in Agriculture and Environment*, Springer, Cham, 2020, https://doi.org/10.1007/978-3-030-27157-2_5.
- [7] S.E. Jacobsen, The worldwide potential for quinoa (*Chenopodium quinoa* Willd.), *Food Rev. Int.* 19 (2003) 167–177, <https://doi.org/10.1081/FRI-120018883>.
- [8] K.B. Ruiz, S. Biondi, R. Oses, et al., Quinoa biodiversity and sustainability for food security under climate change. A review, *Agron. Sustain. Dev.* 34 (2014) 349–359, <https://doi.org/10.1007/s13593-013-0195-0>.
- [9] R. Choukr-Allah, N.K. Rao, A. Hirich, M. Shahid, A. Alshankiti, K. Toderich, S. Gill, K.U.R. Butt, Quinoa for marginal environments: toward future food and nutritional security in MENA and Central Asia regions, *Front. Plant Sci.* 7 (2016) 346, <https://doi.org/10.3389/fpls.2016.00346>.
- [10] H.D. Bertero, A.J. de la Vega, G. Correa, S.E. Jacobsen, A. Mujica, Genotype and genotype-by-environment interaction effects for grain yield and grain size of quinoa (*Chenopodium quinoa* Willd.) as revealed by pattern analysis of international multi-environment trials, *Field Crops Res.* 89 (2004) 299–318, <https://doi.org/10.1016/j.fcr.2004.02.006>.
- [11] A. Bhargava, S. Shukla, D. Ohri, Genetic variability and interrelationship among various morphological and quality traits in quinoa (*Chenopodium quinoa* Willd.), *Field Crops Res.* 101 (2007) 104–116, <https://doi.org/10.1016/j.fcr.2006.10.001>.
- [12] M.I. Hussain, A. Muscolo, M. Ahmed, M.A. Asghar, A.J. Al-Dakheel, Agromorphological, yield and quality traits and interrelationship with yield stability in quinoa (*Chenopodium quinoa* Willd.) genotypes under saline marginal environment, *Plants* 9 (2020) 1763, <https://doi.org/10.3390/plants9121763>.
- [13] L. Wan, H. Cen, J. Zhu, J. Zhang, Y. Zhu, D. Sun, X. Du, L. Zhai, H. Weng, Y. Li, X. Li, Y. Bao, J. Shou, Y. He, Grain yield prediction of rice using multi-temporal UAV-based RGB and multispectral images and model transfer – a case study of small farmlands in the South of China, *Agric. For. Meteorol.* 291 (2020) 108096, <https://doi.org/10.1016/j.agrformet.2020.108096>.
- [14] P. Killeen, L. Kiringa, T. Yeap, P. Branco, Corn grain yield prediction using UAV-based high spatiotemporal resolution imagery, machine learning, and spatial cross-validation, *Remote Sens.* 16 (2024) 683, <https://doi.org/10.3390/rs16040683>.
- [15] Y. Ge, Z. Zhu, S. Jin, J. Zang, R. Zhang, Q. Li, Z. Sun, S. Liu, H. Xu, Z. Zhai, Winter wheat yield prediction using UAV-based multivariate time series data and variate-independent tokenization, *Plant Phenomics* 7 (2025) 100039, <https://doi.org/10.1016/j.plaphe.2025.100039>.
- [16] W.H. Maes, K. Steppe, Perspectives for remote sensing with unmanned aerial vehicles in precision agriculture, *Trends Plant Sci.* 24 (2019) 152–164, <https://doi.org/10.1016/j.tplants.2018.11.007>.
- [17] D.C. Tsouros, S. Bibi, P.G. Sarigiannidis, A review on UAV-based applications for precision agriculture, *Information* 10 (2019) 349, <https://doi.org/10.3390/info10110349>.
- [18] J. Bendig, A. Bolten, S. Bennertz, J. Broscheit, S. Eichfuss, G. Bareth, Estimating biomass of barley using crop surface models (CSMs) derived from UAV-based RGB imaging, *Remote Sens.* 6 (2014) 10395–10412, <https://doi.org/10.3390/rs61110395>.
- [19] F.H. Holman, A.B. Riche, A. Michalski, M. Castle, M.J. Wooster, M.J. Hawkesford, High throughput field phenotyping of wheat plant height and growth rate in field plot trials using UAV based remote sensing, *Remote Sens.* 8 (2016) 1031, <https://doi.org/10.3390/rs8121031>.
- [20] S. Sesay, T. Yamaguchi, S. Kushino, Y. Yoshikawa, S. Adachi, K. Katsura, Fusion of UAV-based 3D mesh and spectral features improves quinoa biomass and LAI estimation across genotypic and temporal variations, *Smart Agric. Technol.* 10 (2025) 100818, <https://doi.org/10.1016/j.atech.2025.100818>.
- [21] K.G. Liakos, P. Busato, D. Moshou, S. Pearson, D. Bochtis, Machine learning in agriculture: a review, *Sensors* 18 (2018) 2674, <https://doi.org/10.3390/s18082674>.
- [22] J.H. Jeong, J.P. Resop, N.D. Mueller, D.H. Fleisher, K. Yun, E.E. Butler, et al., Random forests for global and regional crop yield predictions, *PLoS ONE* 11 (2016) e0156571, <https://doi.org/10.1371/journal.pone.0156571>.
- [23] A. Abdulhamedh, Comparison between common statistical modeling techniques used in research, including: discriminant analysis vs logistic regression, ridge regression vs LASSO, and decision tree vs random forest, *Open Access Libr. J.* 9 (2022) e8414, <https://doi.org/10.4236/oalib.1108414>.
- [24] J. Jiang, K. Johansen, C.S. Stanschewski, H.M. Migdadi, Phenotyping a diversity panel of quinoa using UAV-retrieved leaf area index, SPAD-based chlorophyll and a random forest approach, *Precision Agric.* 23 (2022) 961–983, <https://doi.org/10.1007/s11119-021-09870-3>.

- [25] S. Sankaran, C.Z. Espinoza, L. Hinojosa, X. Ma, K. Murphy, High-throughput field phenotyping to assess irrigation treatment effects in quinoa, *Agrosyst. Geosci. Environ.* 2 (2019) 180063, <https://doi.org/10.2134/age2018.12.0063>.
- [26] K. Ivushkin, H. Bartholomeus, A.K. Bregt, A. Pulatov, M.H.D. Franceschini, H. Kramer, E.N. van Loo, V. Jaramillo Roman, R. Finkers, UAV based soil salinity assessment of cropland, *Geoderma* 338 (2019) 502–512, <https://doi.org/10.1016/j.geoderma.2018.09.046>.
- [27] M. Maimaitijiang, V. Sagan, P. Sidike, S. Hartling, F. Esposito, F.B. Fritsch, Soybean yield prediction from UAV using multimodal data fusion and deep learning, *Remote Sens. Environ.* 237 (2020) 111599, <https://doi.org/10.1016/j.rse.2019.111599>.
- [28] L. Zhou, D. Han, G. Sun, Y. Liu, X. Yan, H. Jia, L. Yan, P. Feng, Y. Li, L. Qiu, Y. Ma, Soybean yield estimation and lodging discrimination based on lightweight UAV and point cloud deep learning, *Plant Phenomics* 7 (2025) 100028, <https://doi.org/10.1016/j.plaphe.2025.100028>.
- [29] W. Xu, P. Chen, Y. Zhan, S. Chen, L. Zhang, Y. Lan, Cotton yield estimation model based on machine learning using time series UAV remote sensing data, *Int. J. Appl. Earth Obs. Geoinf.* 104 (2021) 102511, <https://doi.org/10.1016/j.jag.2021.102511>.
- [30] S. Yang, L. Li, S. Fei, M. Yang, Z. Tao, Y. Meng, Y. Xiao, Wheat yield prediction using machine learning method based on UAV remote sensing data, *Drones* 8 (2024) 284, <https://doi.org/10.3390/drones8100559>.
- [31] J. Yuan, Y. Zhang, Z. Zheng, W. Yao, W. Wang, L. Guo, Grain crop yield prediction using machine learning based on UAV remote sensing: a systematic literature review, *Drones* 8 (2024) 559, <https://doi.org/10.3390/drones8100559>.
- [32] J. Quille-Mamani, L. Ramos-Fernández, J. Huanqueño-Murillo, D. Quispe-Tito, L. Cruz-Villacorta, E. Pino-Vargas, L. Flores del Pino, E. Heros-Aguilar, L.A. ngel Ruiz, Rice yield prediction using spectral and textural indices derived from UAV imagery and machine learning models in Lambayeque, Peru, *Remote Sens.* 17 (2025) 632, <https://doi.org/10.3390/rs17040632>.
- [33] S. Geerts, D. Raes, M. Garcia, R. Miranda, J.A. Cusicanqui, C. Taboada, J. Mendoza, R. Huanca, A. Mamani, O. Condori, J. Mamani, B. Morales, V. Osco, P. Steduto, Simulating yield response of quinoa to water availability with AquaCrop, *Agron. J.* 101 (2009) 499–508, <https://doi.org/10.2134/agronj2008.0137s>.
- [34] C. Pulvento, M. Riccardi, A. Lavini, R. D'andria, R. Ragab, SALTMED model to simulate yield and dry matter for quinoa crop and soil moisture content under different irrigation strategies in South Italy, *Irrig. Drain.* 62 (2013) 229–238, <https://doi.org/10.1002/ird.1727>.
- [35] F. Kaoutar, H. Abdelaziz, B. Ouafe, C.-A. Redouane, R. Ragab, Yield and dry matter simulation using the Sautmed model for five quinoa (*Chenopodium quinoa*) accessions under deficit irrigation in South Morocco, *Irrig. Drain.* 66 (2017) 340–350, <https://doi.org/10.1002/ird.2116>.
- [36] F. Razzaghi, F. Plauborg, S.-E. Jacobsen, C.R. Jensen, M.N. Andersen, Effect of nitrogen and water availability of three soil types on yield, radiation use efficiency and evapotranspiration in field-grown quinoa, *Agric. Water Manage.* 109 (2012) 20–29, <https://doi.org/10.1016/j.agwat.2012.02.002>.
- [37] A. Hirich, R. Choukr-Allah, S.-E. Jacobsen, Quinoa in Morocco – effect of sowing dates on development and yield, *J. Agron. Crop Sci.* 200 (2014) 371–377, <https://doi.org/10.1111/jac.12071>.
- [38] N. Wang, F. Wang, C.C. Shock, C. Meng, L. Qiao, Effects of management practices on quinoa growth, seed yield, and quality, *Agronomy* 10 (2020) 445, <https://doi.org/10.3390/agronomy10030445>.
- [39] C.F. Jordan, Derivation of leaf-area index from quality of light on the forest floor, *Ecology* 50 (1969) 663–666, <https://doi.org/10.2307/1936256>.
- [40] G.S. Birth, G. McVey, Measuring the color of growing turf with a reflectance spectroradiometer, *Agron. J.* 60 (1968) 640–643, <https://doi.org/10.2134/agronj1968.0002196200600060016x>.
- [41] J.-L. Roujean, F.-M. Breon, Estimating PAR absorbed by vegetation from bi-directional reflectance measurements, *Remote Sens. Environ.* 51 (1995) 375–384, [https://doi.org/10.1016/0034-4257\(94\)00114-3](https://doi.org/10.1016/0034-4257(94)00114-3).
- [42] J. Qi, A. Chehbouni, A.R. Huete, Y.H. Kerr, S. Sorooshian, A modified soil adjusted vegetation index, *Remote Sens. Environ.* 48 (1994) 119–126, [https://doi.org/10.1016/0034-4257\(94\)90134-1](https://doi.org/10.1016/0034-4257(94)90134-1).
- [43] G. Rondeaux, M. Steven, F. Baret, Optimization of soil-adjusted vegetation indices, *Remote Sens. Environ.* 55 (1996) 95–107, [https://doi.org/10.1016/0034-4257\(95\)00186-7](https://doi.org/10.1016/0034-4257(95)00186-7).
- [44] A. Huete, K. Didan, T. Miura, E.P. Rodriguez, X. Gao, L.G. Ferreira, Overview of the radiometric and biophysical performance of the MODIS vegetation indices, *Remote Sens. Environ.* 83 (2002) 195–213, [https://doi.org/10.1016/s0034-4257\(02\)00096-2](https://doi.org/10.1016/s0034-4257(02)00096-2).
- [45] J.W. Rouse, R.W. Haas, J.A. Schell, D.W. Deering, J.C. Harlan, Monitoring the vertical advancements (greenwave effect) and retrogradation of natural vegetation. NASA/GSFC Type III Final Report, Texas A & M University, Remote Sensing Center, College Station, 1974, pp. 1–137.
- [46] A.R. Huete, A soil-adjusted vegetation index (SAVI), *Remote Sens. Environ.* 25 (1988) 295–309, [https://doi.org/10.1016/0034-4257\(88\)90106-X](https://doi.org/10.1016/0034-4257(88)90106-X).
- [47] M. Ali, C. Montzka, A. Stadler, G. Menz, F. Thonfeld, H. Vereecken, Estimation and validation of RapidEye-based time-series of leaf area index for winter wheat in the Ruhr catchment (Germany), *Remote Sens.* 7 (2015) 2808–2831, <https://doi.org/10.3390/rs70302808>.
- [48] Q. Xie, J. Dash, W. Huang, D. Peng, Q. Qin, H. Mortimer, et al., Vegetation indices combining the red and red-edge spectral information for leaf area index retrieval, *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* 11 (2018) 1482–1493, <https://doi.org/10.1109/JSTARS.2018.2813281>.
- [49] D.A. Sims, J.A. Gamon, Relationships between leaf pigment content and spectral reflectance across a wide range of species, leaf structures and developmental stages, *Remote Sens. Environ.* 81 (2002) 337–354, [https://doi.org/10.1016/S0034-4257\(02\)00010-X](https://doi.org/10.1016/S0034-4257(02)00010-X).
- [50] G.A. Carter, Ratios of leaf reflectances in narrow wavebands as indicators of plant stress, *Int. J. Remote Sens.* 15 (1994) 697–703, <https://doi.org/10.1080/01431169408954109>.
- [51] A.E. Hoerl, R.W. Kennard, Ridge regression: biased estimation for nonorthogonal problems, *Technometrics* 12 (1970) 55–67, <https://doi.org/10.2307/1267351>.
- [52] T. Cover, P. Hart, Nearest neighbor pattern classification, *IEEE Trans. Inf. Theory* 13 (1967) 21–27, <https://doi.org/10.1109/TIT.1967.1053964>.
- [53] L. Breiman, Random forests, *Mach. Learn.* 45 (2001) 5–32, <https://doi.org/10.1023/A:1010933404324>.
- [54] T. Chen, C. Guestrin, XGBoost: a scalable tree boosting system, in: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785–794, <https://doi.org/10.1145/2939672.2939785>.
- [55] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, T.-Y. Liu, LightGBM: a highly efficient gradient boosting decision tree, *Adv. Neural Inf. Process. Syst.* 30 (2017).
- [56] S.M. Lundberg, S.I. Lee, A unified approach to interpreting model predictions, *Adv. Neural Inf. Process. Syst.* 30 (2017).
- [57] D. Gómez, P. Salvador, J. Sanz, J.L. Casanova, Modelling wheat yield with antecedent information, satellite and climate data using machine learning methods in Mexico, *Agric. For. Meteorol.* 300 (2021) 108317, <https://doi.org/10.1016/j.agrformet.2020.108317>.
- [58] M. Cheng, J. Penuelas, M.F. McCabe, C. Atzberger, X. Jiao, W. Wu, X. Jin, Combining multi-indicators with machine-learning algorithms for maize yield early prediction at the county-level in China, *Agric. For. Meteorol.* 323 (2022) 109057, <https://doi.org/10.1016/j.agrformet.2022.109057>.
- [59] H. Ge, Q. Zhang, M. Shen, Y. Qin, L. Wang, C. Yuan, Enhancing yield prediction in maize breeding using UAV-derived RGB imagery: a novel classification-integrated regression approach, *Front. Plant Sci.* 16 (2025) 1511871, <https://doi.org/10.3389/fpls.2025.1511871>.
- [60] J. Pei, S. Tan, Y. Zou, C. Liao, Y. He, J. Wang, H. Huang, T. Wang, H. Tian, H. Fang, L. Wang, J. Huang, The role of phenology in crop yield prediction: comparison of ground-based phenology and remotely sensed phenology, *Agric. For. Meteorol.* 361 (2025) 110340, <https://doi.org/10.1016/j.agrformet.2024.110340>.
- [61] X. Zhou, H.B. Zheng, X.Q. Xu, J.Y. He, X.K. Ge, X. Yao, T. Cheng, Y. Zhu, W.X. Cao, Y.C. Tian, Predicting grain yield in rice using multi-temporal vegetation indices from UAV-based multispectral and digital imagery, *ISPRS J. Photogramm. Remote Sens.* 130 (2017) 246–255, <https://doi.org/10.1016/j.isprsjprs.2017.05.003>.
- [62] Z. Lou, X. Lu, S. Li, Yield prediction of winter wheat at different growth stages based on machine learning, *Agronomy* 14 (2024) 1834, <https://doi.org/10.3390/agronomy14081834>.
- [63] P. Zhang, B. Lu, J. Shang, X. Wang, Z. Hou, S. Jin, Y. Yang, H. Zang, J. Ge, Z. Zeng, Ensemble learning for oat yield prediction using multi-growth stage UAV images, *Remote Sens.* 16 (2024) 4575, <https://doi.org/10.3390/rs16234575>.
- [64] H.D. Bertero, R.A. Ruiz, Determination of seed number in sea level quinoa (*Chenopodium quinoa* Willd.) cultivars, *Eur. J. Agron.* 28 (2008) 186–194, <https://doi.org/10.1016/j.eja.2007.07.002>.
- [65] S.C. Vázquez, A. del Pozo, D. Castillo, I. Matus, L. Gómez-Pando, D. Zamudio-Ayala, C.M. Mignone, H.D. Bertero, D.F. Calderini, The critical period for yield and grain protein determination in quinoa (*Chenopodium quinoa* Willd.), *Field Crops Res.* 306 (2024) 109207, <https://doi.org/10.1016/j.fcr.2023.109207>.
- [66] J.A. Gibbs, M. Pound, A.P. French, D.M. Wells, E. Murchie, T. Pridmore, Approaches to three-dimensional reconstruction of plant shoot topology and geometry, *Funct. Plant Biol.* 44 (2016) 62–75, <https://doi.org/10.1071/FP16167>.
- [67] M. Maimaitijiang, V. Sagan, P. Sidike, M. Maimaitiyiming, S. Hartling, K. T. Peterson, M. Maw, N. Shakoar, T. Mockler, F.B. Fritsch, Vegetation index weighted canopy volume model (CVMVI) for soybean biomass estimation from unmanned aerial system-based RGB imagery, *ISPRS J. Photogramm. Remote Sens.* 151 (2019) 27–41, <https://doi.org/10.1016/j.isprsjprs.2019.03.003>.
- [68] D. Dimov, P. Noack, Exploring the potential of multi-temporal crop canopy models and vegetation indices from Pleiades imagery for yield estimation, *Remote Sens.* 15 (2023) 3990, <https://doi.org/10.3390/rs15163990>.
- [69] D. Kim, W.H. Kang, I. Hwang, J. Kim, J.H. Kim, K.S. Park, J.E. Son, Use of structurally-accurate 3D plant models for estimating light interception and photosynthesis of sweet pepper (*Capsicum annuum*) plants, *Comput. Electron. Agric.* 177 (2020) 105689, <https://doi.org/10.1016/j.compag.2020.105689>.
- [70] A.J. de la Vega, M.A. Cantore, M.M. Sposaro, N. Trapani, M.López Pereira, A. J. Hall, Canopy stay-green and yield in non-stressed sunflower, *Field Crops Res.* 121 (2011) 175–185, <https://doi.org/10.1016/j.fcr.2010.12.015>.
- [71] H. Thomas, H. Ougham, The stay-green trait, *J. Exp. Bot.* 65 (2014) 3889–3900, <https://doi.org/10.1093/jxb/eru037>.
- [72] F. Liu, Q. Song, J. Zhao, L. Mao, H. Bu, Y. Hu, X.-G. Zhu, Canopy occupation volume as an indicator of canopy photosynthetic capacity, *New Phytol.* 232 (2021) 941–956, <https://doi.org/10.1111/nph.17611>.
- [73] S. Xiao, S. Fei, Q. Li, B. Zhang, H. Chen, D. Xu, Z. Cai, K. Bi, Y. Guo, B. Li, The importance of using realistic 3D canopy models to calculate light interception in the field, *Plant Phenomics* 5 (2023) 0082, <https://doi.org/10.34133/plantphenomics.0082>.
- [74] A. Šupčík, G. Milics, I. Matečnjak, Predicting grape yield with vine canopy morphology analysis from 3D point clouds generated by UAV imagery, *Drones* 8 (2024) 216, <https://doi.org/10.3390/drones8050216>.
- [75] T. Satapathy, D. Dutta, Characterizing the vertical variability of canopy structure from space: implications for biodiversity, productivity, and ecosystem functioning, *Agric. For. Meteorol.* 372 (2025) 110710, <https://doi.org/10.1016/j.agrformet.2025.110710>.